Incentives in Networks for Congestion Control

Nirav B Mehta
Computer Science Department, Stanford University, Stanford CA 94305
niravbm@cs.stanford.edu

ABSTRACT

This paper analyses the problem of congestion in the internet and analyses game theoretic principle of providing incentives to the users to control congestion. Various pricing mechanisms to achieve differential Quality of Service (QoS) are also analyzed. An overview of the techniques presented in current literature to make congestion control incentive compatible for the users of the internet.

1. Introduction

Congestion has been the bane of the internet since its very inception. Even though bandwidth has kept on increasing at a high rate, user demand for bandwidth has always outstripped the available supply, with a variety of high-bandwidth applications being ushered in everyday. The result is congestion.

2. Why the TCP Congestion Control Algorithm may not control congestion?

Congestion control in TCP takes place by means of a *backoff mechanism.* If the rate at which you are sending packets causes congestion on the network, the source TCP follows the exponential backoff algorithm and reduces the packet flow onto the network. The TCP then gradually increases the sending rate till the next indication of congestion. But this mechanism is not incentive compatible; users can modify their TCP so that it keeps on sending packets even in the face of congestion. This way the user with such a perverted TCP will enjoy maximum utilization of the network resources even during congestion as other TCP's would have backed off. If the number of such perverted TCP versions increases congestion will become a severe problem. TCP congestion control algorithm will not achieve the aim of controlling congestion in this situation.

In the next few sections we analyze a number of techniques in current literature to address this problem. We start of with a discussion of the Paris Metro Pricing (PMP) by Odlyzko, followed by a technique of resource pricing presented by Gibbens and Kelly. We then analyze the experimental work of Key and McAuley. As an aside there is also an overview of Leyton-Brown et alls' work on smoothing demand for focused loads. We finally conclude with a comparison of the various techniques along with some pointers for future work.

3. Paris Metro Pricing (PMP)

Odlyzko proposes a logical partitioning of the internet, with each partition having a different price for the users. The partitions are equivalent, the only guarantee being that higher priced networks will attract lesser traffic and hence will provide a higher quality of service (QoS) The

technique plans to make use of the 3-bit Priority field in the IP header to identify the destination logical network for each packet.
Thus the user can decide what price he wants to pay by selecting the network to use.

Critique:

- Use of a 3-bit priority field may restrict the number of logical networks to 8. With more and more diverse applications coming in everyday, a need for distinguishing the internet into more than 8 logical networks will soon become a necessity.
- All the current applications using the internet currently will have to be rewritten to make a selection of the network to use based on the user decision.
- All Usage sensitive pricing mechanisms such as this require additional hardware and software to count the packets or bytes per user, to charge the user for his usage.
- Nevertheless this technique is quite appealing because of its simplicity and self regulating mechanism.

4. Resource Pricing (RP)

Similar to the PMP discussed above, the technique of resource pricing termed *proportionally fair pricing* described by Gibbens and Kelly is also quite simple. It charges users a fixed-price per marked packet but the difference is that the cost distinction here is made by the network and not the user. The argument put forward by them is that it is easier to achieve an efficient allocation by conveying information on congestion from the network to intelligent end nodes rather than by requiring users to classify their flows into predefined categories and conveying this information from the users to the network.

4.1 Marking Packets at Resources

Marking of packets is done at the resources. Any packet that arrives at a resource during the "critical congestion interval" (the interval between the start of a busy period and a packet loss within the same busy period at a resource) is marked. The marking is done on the premise that any packet arriving at the resource during an overloaded slot adds to the congestion, hence should be marked.
When the end nodes receive information about the marked packets at the overloaded resource, they can take steps to control the congestion. This works fine in a network with co-operative end nodes, otherwise a small positive charge may be applied per marked packet to ensure that the end-nodes also have incentive to use the network efficiently.

Critique:

- Maintaining a virtual resource shadow within the actual resource and then marking based on congestion in the virtual resource results in an early warning of congestion.
- In the internet, congestion is signaled by dropping of packets, but this is a very extreme strategy compared to the marking strategy where the end nodes get a notification of congestion through marked nodes, the proportion of dropped nodes is very low.
- As an aside, in the current TCP algorithm due to Jacobson the source halves the congestion window size in response to congestion signals from the internet. At the same time the number of congestion signals received by a flow is roughly proportional to the size of the flow. This results in 2 multiplicative effects while reducing the flow, while in the algorithm proposed by Gibbens and Kelly the source reduces the flow by a constant amount in response to a congestion indication signal resulting in an additive as opposed to multiplicative reduction. The resources on the other hand produce congestion

indication signals at a rate proportional to the size of the flow, which when combined with the additive reduction by the source ensures that the overall decrease in rate is multiplicative. This therefore results in better utilization of the network, in the period immediately succeeding the congested period, where otherwise the flow rate would have dropped quite low.

- Another point to be noted is that the source need not start from the low initial rate every time it connects to the internet; if the network is not congested, the source may start directly at a high rate again resulting in more efficient use of the network. By means of the marking mechanism the end nodes can decide for themselves as they have information about the network through previous experiences of using the network.

4.2 Assessing User Strategies

Complementary to the marking of packets for resource pricing, Gibbens and Key have performed experimental analysis of the sort of strategies that users may adopt, given the framework of the internet as in 4.1. This work concentrates more on the strategies that users may adopt given that they are charged for each marked acknowledgement that they receive.

The user's objective is to use the network to accomplish a particular task (e.g. transfer a file) in time T, at minimum cost. Here as the number of marks received denotes cost to the user, the aim of the user is to minimize the number of marks received, by adjusting the flow rate onto the network.

The 4 strategies that are analyzed to achieve this are:

1. *Constant Rate* – The user distributes the data to be transferred uniformly over the time interval T.
2. *Last One* – Packets are sent at a high rate(xh) if the previous acknowledgement received was unmarked else a low rate(xl) is used.
3. *Last Two* – Similar to the last one, only it considers the last ACK's received.
4. *Estimator* – This tries to derive an estimate of the current congestion in the network based on the proportion of marked ACK's received so far, and accordingly adjusts the rate of flow.

Significantly the Last one strategy performed better in most of the experiments, suggesting that short term fluctuations in the marking rate can be exploited, analogous to the winning of the Tit-for-Tat strategy in Axelrod's Prisoners Dilemma competition.

Critique:

- As users are allowed to freely select their flow rates, differential QoS is implemented implicitly.
- Testing of such a prototype network can be done by running a distributed emulation of a network that is a multi-user distributed game, where the users can come up with any scheme that they desire to maximize transfer at minimum cost.

5. Incentives for tackling focused loads:

In most networks, if we observe the average utilization of network capacity, it is well below the maximum. But even then congestion occurs. The reason is short periods during which the load is highly focused so that it overloads the network resources. Leyton-Brown et all propose four mechanisms to provide incentives to the users to spread out their load temporally.

The basic principle used by this technique is to make it more attractive for users to depart from their true preferences by probabilistically waiving off the cost of network resource usage in some time slot.

Critique:

- In many networks focused load may be the only form of congestion, and tackling it as above may completely solve the problem of congestion obviating the need for expensive technological mechanisms.
- Such a mechanism is bound to increase the load balancing capability of usage sensitive price mechanisms (for example PMP).
- On the other hand, some of the mechanisms discussed like the Bulletin board for example incur high overhead in terms of time cost for communication and coordination.
- At the same time, it may not be always possible to provide incentives to users to change usage time, for example users have no incentive to avoid accessing web servers close to deadlines; hence this technique is more of a resource specific solution rather than a generic solution.

6. PMP versus RP:

	PMP	**RP**
Price of a packet decided by:	User	Network
Logical partitioning of the internet	Yes	A single network
Flexibility to users	Less flexible as users can choose 1 network from the available categories.	Flexible as users can constantly change their flow rate depending on requirements.
Marking Mechanism	Not required	Each bit arriving at a resource in an overloaded slot needs to be marked; each marked bit has to be accounted to determine cost to user.
How is congestion controlled?	If congestion occurs, some of the users may switch to high price networks, relieving congestion.	If congestion occurs, some users will reduce their flow rate to minimize the number of their marked packets.
Changes to current Applications	Rewriting of all applications that use the internet will be required so that the user is given the selection of the network to use from within the application.	A number of user strategies as discussed in 4.2 will evolve, users can select any of those for use in their TCP, and applications need not be modified.

Both PMP and RP need to implement in either hardware or software, a means to count the packets (marked packets in RP) per user, to determine the charge to each user.

7. Pointers to Future Work

A number of questions on feasibility of these techniques are yet unanswered, like:

- How do you interface between networks that use usage sensitive pricing and those that do not?
- How will revenue splitting occur between the different service providers?
- Such techniques will change the way that users surf the net, hence E-commerce sites will need a mechanism analogous to the 800 in telephones where the receiver can pay for the traffic generated. This is a significant issue to be tackled.
- Complete congestion control is still not assured. For example, in the PMP mechanism if one of the lower priced networks starts experiencing congestion, a large number of users may migrate to higher priced networks thereby congesting that network too. Hence the problem has in effect just shifted from one network to another.

8. References

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